Lines, Triangles & Polygons

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Parallel lines and transversals**

Two parallel lines are cut by a transversal, forming an angle of 105° on one line. Find the corresponding angle, the alternate interior angle, and the co-interior (same-side interior) angle on the other line.

Corresponding angle $= 105^\circ$ (equal). Alternate interior angle $= 105^\circ$ (equal). Co-interior angle $= 180^\circ - 105^\circ = 75^\circ$ (supplementary).

2 **[Foundational]** • **Target: Parallel lines and transversals**

Two angles formed by a transversal crossing parallel lines are alternate exterior angles, so they are equal. If one measures $3x + 10$ and the other measures $5x - 30$, find x and the measure of the angle.

$3x + 10 = 5x - 30 \Rightarrow 40 = 2x \Rightarrow x = 20$. The angle measures $3(20) + 10 = 70^\circ$.

3 **[Foundational]** • **Target: Parallel lines and transversals**

Two parallel lines are cut by a transversal. One interior angle is $4x + 15$, and its corresponding angle on the other line is $6x - 25$. Find x and the angle measure.

Corresponding angles are equal: $4x + 15 = 6x - 25 \Rightarrow 2x = 40 \Rightarrow x = 20$. The angle measures $4(20) + 15 = 95^\circ$.

4 **[Foundational]** • **Target: Triangle angle sum and the exterior angle theorem**

The angles of a triangle are in the ratio $3 : 5 : 7$. Find each angle.

The parts sum to $3 + 5 + 7 = 15$, and the total is 180°, so each part is 12°. The angles are 36°, 60°, and 84°.

5 **[Foundational]** • **Target: Triangle angle sum and the exterior angle theorem**

In triangle ABC, $\angle A = 52^\circ$ and the exterior angle at C is 130°. Find $\angle B$.

The exterior angle equals the sum of the remote interior angles: $130^\circ = 52^\circ + \angle B \Rightarrow \angle B = 78^\circ$.

6 **[Foundational]** • **Target: Triangle angle sum and the exterior angle theorem**

An exterior angle of a triangle measures 115°, and one remote interior angle is 50°. Find the other remote interior angle and the third (adjacent) interior angle.

The other remote interior angle is $115^\circ - 50^\circ = 65^\circ$. The adjacent interior angle is supplementary to the exterior angle: $180^\circ - 115^\circ = 65^\circ$. Check: $50^\circ + 65^\circ + 65^\circ = 180^\circ$.

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7 [Foundational] • Target: Triangle congruence criteria

State which congruence criterion (SSS, SAS, ASA, AAS, or RHS) proves triangles PQR and XYZ congruent, given $PQ = XY$, $\angle Q = \angle Y$, and $QR = YZ$.

SAS — two sides and the included angle.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Triangle congruence criteria

Triangle DEF has $DE = 9$, $EF = 12$, $\angle E = 90^\circ$, and triangle GHI has $GH = 9$, $HI = 12$, $\angle H = 90^\circ$. Are the triangles congruent? By which criterion?

Yes — congruent by SAS, since two sides and the included right angle are equal in both triangles.

9 [Intermediate] • Target: Triangle congruence criteria

Two triangles have all three pairs of corresponding sides equal in length. Which congruence criterion applies, and are the triangles necessarily congruent?

SSS applies, and yes, the triangles are necessarily congruent.

10 [Intermediate] • Target: Similar triangles and proportional reasoning

Triangle $ABC \sim$ triangle DEF, with $AB = 6$, $BC = 8$, and $DE = 9$. Find EF.

Scale factor $= \frac{DE}{AB} = \frac{9}{6} = 1.5$. $EF = 8 \times 1.5 = 12$.

11 [Intermediate] • Target: Similar triangles and proportional reasoning

Triangle ABC shown below has $AB = 12$, $BC = 16$, and $CA = 20$. It is similar to triangle DEF, where $DE = 15$ corresponds to AB. Find EF and FD.

Scale factor $= \frac{DE}{AB} = \frac{15}{12} = 1.25$. $EF = BC \times 1.25 = 16 \times 1.25 = 20$. $FD = CA \times 1.25 = 20 \times 1.25 = 25$.

12 [Intermediate] • Target: Similar triangles and proportional reasoning

Triangle JKL has $JK = 15$ and $KL = 21$, and is similar to triangle MNO with scale factor $\frac{2}{3}$ (triangle MNO is smaller), where MN corresponds to JK. Given $MN = 10$, verify the scale factor is consistent, then find NO.

Check: $\frac{MN}{JK} = \frac{10}{15} = \frac{2}{3}$, consistent. $NO = KL \times \frac{2}{3} = 21 \times \frac{2}{3} = 14$.

13 [Intermediate] • Target: The Law of Cosines

In triangle PQR shown below, $PQ = 9$, $QR = 13$, and $\angle Q = 50^\circ$. Find $PR = x$, to the nearest hundredth.

$x^2 = 9^2 + 13^2 - 2(9)(13)\cos 50^\circ \approx 250 - 150.41 = 99.59$, so $x \approx \sqrt{99.59} \approx 9.98$.

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14 [Intermediate] • Target: The Law of Cosines

A triangle has sides $a = 7$, $b = 8$, $c = 10$. Use the Law of Cosines to find angle C (opposite side c), to the nearest tenth of a degree.

$\cos C = \frac{a^2 + b^2 - c^2}{2ab} = \frac{49 + 64 - 100}{112} = \frac{13}{112} \approx 0.1161$. $C \approx 83.3^\circ$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: The Law of Cosines • Cross-Lesson: 5.01 Trigonometry

A triangle has sides $a = 6$, $b = 6$, and included angle $C = 80^\circ$. Find side c, to the nearest hundredth, and state what type of triangle this is based on the two given equal sides.

$c^2 = 6^2 + 6^2 - 2(6)(6)\cos 80^\circ = 72 - 72(0.1736) \approx 59.50$, so $c \approx 7.71$. Since two sides are equal, the triangle is isosceles.

16 [Challenging] • Target: Heron's formula • Cross-Lesson: 6.01 Circles & Coordinate Geometry

Find the area of a triangle with sides 13, 14, 15 using Heron's formula.

$s = \frac{13 + 14 + 15}{2} = 21$. Area $= \sqrt{21(21-13)(21-14)(21-15)} = \sqrt{21 \cdot 8 \cdot 7 \cdot 6} = \sqrt{7056} = 84$.

17 [Challenging] • Target: Heron's formula • Cross-Lesson: 7.01 Solid Geometry & Similarity

A triangular plot of land has sides 9, 10, 17. Find its area using Heron's formula.

$s = \frac{9 + 10 + 17}{2} = 18$. Area $= \sqrt{18(18-9)(18-10)(18-17)} = \sqrt{18 \cdot 9 \cdot 8 \cdot 1} = \sqrt{1296} = 36$.

18 [Challenging] • Target: Heron's formula • Cross-Lesson: 8.01 Statistics & Probability

A triangular sail has sides 5 m, 12 m, and 13 m. Find its area using Heron's formula, and verify using the right-triangle area formula (since 5-12-13 is a Pythagorean triple).

$s = \frac{5 + 12 + 13}{2} = 15$. Area $= \sqrt{15 \cdot 10 \cdot 3 \cdot 2} = \sqrt{900} = 30$ m^2. Check: since it's a right triangle with legs 5 and 12, Area $= \frac{1}{2}(5)(12) = 30$ m^2. Matches.

19 [Challenging] • Target: Interior and exterior angles of polygons • Cross-Lesson: 1.01 Advanced Algebra

Find the sum of the interior angles of a regular decagon (10 sides), and the measure of each interior angle.

Sum $= (10 - 2) \times 180^\circ = 1440^\circ$. Each interior angle $= \frac{1440^\circ}{10} = 144^\circ$.

20 [Challenging] • Target: Interior and exterior angles of polygons • Cross-Lesson: 2.01 Functions: Composite, Inverse & Logarithmic

Each exterior angle of a regular polygon measures 24°. How many sides does the polygon have?

$n = \frac{360^\circ}{24^\circ} = 15$ sides.

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Parallel lines and transversals

An exterior angle of a triangle measures 82°, and one remote interior angle is 30°. Find the other remote interior angle.

a. 52° ✓ **b.** 57°

c. 47° **d.** 30°

The exterior angle equals the sum of the remote interior angles: $82^\circ = 30^\circ + \angle B \Rightarrow \angle B = 52^\circ$.

22 [Foundational] • Target: Triangle angle sum and the exterior angle theorem

A triangle has sides $a = 6$, $b = 11$, and included angle $C = 110^\circ$. Find side c, to the nearest hundredth.

a. 13.72 **b.** 14.72

c. 14.22 ✓ **d.** 15.22

$c^2 = 6^2 + 11^2 - 2(6)(11)\cos 110^\circ \approx 157 + 45.15 \approx 202.15$, so $c \approx 14.22$.

23 [Foundational] • Target: Triangle congruence criteria

Find the sum of the interior angles of a regular polygon with 20 sides, and the measure of each interior angle.

a. $3060, 162$ **b.** $3240, 162$ ✓

c. $3420, 157$ **d.** $3240, 167$

Sum $= (20 - 2) \times 180^\circ = 3240^\circ$. Each interior angle $= \frac{3240^\circ}{20} = 162^\circ$.

24 [Foundational] • Target: Similar triangles and proportional reasoning

The sum of interior angles of a hexagon is:

a. 540° **b.** **720°** ✓

c. 900° **d.** 360°

Applying Lines, Triangles & Polygons principles, $(6 - 2) \times 180 = 720°$ Therefore, the verified correct option is 720°.

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25 [Foundational] • Target: The Law of Cosines

The three angles of a triangle measure $(3x - 5)^\circ$, $(3x - 10)^\circ$, and $(x + 20)^\circ$. Find the measure of the largest angle.

a. **70** ✓

b. 64

c. 62

d. 65

$(3x - 5) + (3x - 10) + (x + 20) = 180 \Rightarrow x = 25$. Angles: 70°, 65°, 45°; largest $= 70^\circ$.

26 [Foundational] • Target: Heron's formula

Lines $\ell_1 \parallel \ell_2$ are cut by a transversal. Two co-interior (same-side interior) angles measure $(4x + 4)^\circ$ and 120°. Solve for x.

a. 21

b. 8

c. 20

d. **14** ✓

Co-interior angles are supplementary: $(4x + 4) + 120 = 180 \Rightarrow x = 14$.

27 [Foundational] • Target: Interior and exterior angles of polygons

Lines $\ell_1 \parallel \ell_2$ cut by a transversal form corresponding angles of $(4x - 3)^\circ$ and $(3x + 4)^\circ$. Find the measure of the angle $(4x - 3)^\circ$.

a. 26

b. 34

c. **25** ✓

d. 20

$4x - 3 = 3x + 4 \Rightarrow x = 7$. Angle $= 4(7) - 3 = 25^\circ$.

28 [Foundational] • Target: Area of polygons

What is the sum of the interior angles of a convex polygon with 4 sides?

a. 367

b. **360** ✓

c. 368

d. 358

Sum $= (n - 2) \times 180^\circ = (4 - 2) \times 180^\circ = 360^\circ$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: A well-designed word problem: a surveyor's triangular lots

The sum of the interior angles of a convex polygon is 2520°. How many sides does the polygon have?

a. 20

b. **16** ✓

c. 19

d. 3

$(n-2) \times 180 = 2520 \Rightarrow n - 2 = 14 \Rightarrow n = 16$.

30 [Intermediate] • Target: Parallel lines and transversals

In the diagram below, line AB is parallel to line CD. Transversal line EF intersects AB at point G and CD at point H. If the measure of angle AGH is 50°, what is the measure of angle GHD?

a. 50°

b. **130°** ✓

c. 40°

d. 90°

When a transversal intersects two parallel lines, consecutive interior angles are supplementary, meaning they add up to 180 degrees. Angle AGH and angle GHC are consecutive interior angles. Therefore, $m\angle AGH + m\angle GHC = 180^\circ$. Since $m\angle AGH = 50^\circ$, we have $50^\circ + m\angle GHC = 180^\circ$. Solving for $m\angle GHC$ gives $m\angle GHC = 180^\circ - 50^\circ = 130^\circ$. Angle GHD and angle GHC are angles on a straight line, forming a linear pair. Thus.

31 [Intermediate] • Target: Triangle angle sum and the exterior angle theorem

In the diagram below, lines AB and CD are parallel. Line EF is a transversal intersecting AB at G and CD at H. If the measure of angle AGH is $(2x + 10)^\circ$ and the measure of angle GHD is $(3x - 5)^\circ$, find the value of x.

a. 15

b. **25** ✓

c. 35

d. 45

Since lines AB and CD are parallel and EF is a transversal, angle AGH and angle GHD are consecutive interior angles. Consecutive interior angles are supplementary, meaning their measures add up to 180°. Therefore, we can set up the equation: $(2x + 10)^\circ + (3x - 5)^\circ = 180^\circ$ Combine like terms: $5x + 5 = 180$ Subtract 5 from both sides: $5x = 175$ Divide by 5: $x = \frac{175}{5}$ $x = 35$ Let's check the angles: Angle AGH = $2(35) + 10 = 70 + 10 = 80^\circ$ Angle GHD =.

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32 [Intermediate] • Target: Triangle congruence criteria

In the diagram below, triangle ABC is a right-angled triangle with the right angle at C. An altitude is drawn from C to the hypotenuse AB, meeting AB at point D. If AC = 9 and BC = 12, find the length of the altitude CD.

a. **7.2** ✓

b. 10.8

c. 8.4

d. 9.6

Let CD be the altitude from C to the hypotenuse AB in the right-angled triangle ABC. We are given AC = 9 and BC = 12. First, we find the length of the hypotenuse AB using the Pythagorean theorem: $AB^2 = AC^2 + BC^2 = 9^2 + 12^2 = 81 + 144 = 225$. Thus, $AB = \sqrt{225} = 15$. The area of triangle ABC can be calculated in two ways: $\frac{1}{2} \times AC \times BC$ and $\frac{1}{2} \times AB \times CD$. Equating these two expressions for the area, we get.

33 [Intermediate] • Target: Similar triangles and proportional reasoning

In the diagram below, triangle ABC is a right-angled triangle with the right angle at C. CD is the altitude to the hypotenuse AB. If AD = 4 units and DB = 9 units, find the length of the altitude CD.

a. **6 units** ✓

b. 13 units

c. 36 units

d. 7.5 units

In a right-angled triangle, the altitude drawn to the hypotenuse creates three similar triangles: the original triangle and two smaller triangles. Specifically, triangle ADC is similar to triangle CDB. A property derived from this similarity is that the square of the length of the altitude to the hypotenuse is equal to the product of the lengths of the two segments it divides the hypotenuse into. This is known as the Geometric Mean Theorem. Therefore, $CD^2 = AD \times DB$. Given $AD = 4$ and $DB = 9$, we have.

34 [Intermediate] • Target: The Law of Cosines

Find the measure of each interior angle of a regular polygon with 18 sides.

a. 169

b. 163

c. 162

d. **160** ✓

Each interior angle $= \frac{(n-2) \times 180^\circ}{n} = \frac{(18-2) \times 180^\circ}{18} = 160^\circ$.

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35 [Intermediate] • Target: Heron's formula

Each interior angle of a regular polygon measures 120°. How many sides does the polygon have?

a. 12

b. 4

c. **6** ✓

d. 14

$\frac{(n-2)180}{n} = 120 \Rightarrow 180n - 360 = 120n \Rightarrow n(60) = 360 \Rightarrow n = 6.$

36 [Intermediate] • Target: Interior and exterior angles of polygons

Angles S and T are supplementary. If $\angle S = 34^\circ$, find $\angle T$.

a. 139

b. 143

c. 145

d. **146** ✓

Supplementary angles sum to 180°: $\angle T = 180^\circ - 34^\circ = 146^\circ$.

37 [Intermediate] • Target: Area of polygons

Angles M and N are complementary. If $\angle M = 69^\circ$, find $\angle N$.

a. **21** ✓

b. 25

c. 16

d. 18

Complementary angles sum to 90°: $\angle N = 90^\circ - 69^\circ = 21^\circ$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: A well-designed word problem: a surveyor's triangular lots • Cross-Lesson: 8.01 Statistics & Probability

Two lines intersect, forming vertical angles. If one angle measures 114°, what is the measure of its vertical angle?

a. 116

b. 120

c. 119

d. **114** ✓

Vertical angles are congruent, so the vertical angle also measures 114°.

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39 [Challenging] • Target: Parallel lines and transversals • Cross-Lesson: 1.01 Advanced Algebra

Find the measure of each exterior angle of a regular polygon with 12 sides.

a. 36

b. **30** ✓

c. 32

d. 21

Each exterior angle $= \frac{360^\circ}{n} = \frac{360^\circ}{12} = 30^\circ$.

40 [Challenging] • Target: Triangle angle sum and the exterior angle theorem • Cross-Lesson: 2.01 Functions: Composite, Inverse & Logarithmic

Lines $\ell_1 \parallel \ell_2$ are cut by a transversal. One angle measures $(3x + 5)^\circ$ and its corresponding angle measures 80°. Solve for x.

a. 33

b. 20

c. **25** ✓

d. 19

Corresponding angles are equal: $3x + 5 = 80 \Rightarrow x = 25$.

41 [Challenging] • Target: Triangle congruence criteria • Cross-Lesson: 3.01 Sequences, Series & Complex Numbers

A trapezoid has parallel sides 17 and 5, and height 12. Find its area.

a. 134

b. 140

c. **132** ✓

d. 136

Area $= \frac{(b_1 + b_2)h}{2} = \frac{(17 + 5)(12)}{2} = 132$.

42 [Challenging] • Target: Similar triangles and proportional reasoning • Cross-Lesson: 5.01 Trigonometry

A regular polygon has an exterior angle of 36°. How many sides does it have?

a. 6

b. **10** ✓

c. 7

d. 4

$n = \frac{360^\circ}{36^\circ} = 10$.

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43 [Challenging] • Target: The Law of Cosines • Cross-Lesson: 6.01 Circles & Coordinate Geometry

A right triangle has legs 5 and 12. Hypotenuse is:

a. 11

b. **13** ✓

c. 17

d. 15

Applying Lines, Triangles & Polygons principles, $\sqrt{25 + 144} = \sqrt{169} = 13$ Therefore, the verified correct option is 13.

44 [Challenging] • Target: Heron's formula • Cross-Lesson: 7.01 Solid Geometry & Similarity

In the diagram below, line segment DE is parallel to BC. Given that AD = 6, DB = 4, and DE = 12, find the length of BC.

a. 18

b. **20** ✓

c. 30

d. 15

Since DE is parallel to BC, triangle ADE is similar to triangle ABC (AA similarity: $\angle DAE = \angle BAC$ as they are the same angle, and $\angle ADE = \angle ABC$ as corresponding angles when DE || BC and AB is a transversal). Because the triangles are similar, the ratio of corresponding sides is equal: $\frac{AD}{AB} = \frac{AE}{AC} = \frac{DE}{BC}$ We are given AD = 6 and DB = 4. Therefore, AB = AD + DB = 6 + 4 = 10. Using the ratio of similarity: $\frac{AD}{AB} = \frac{DE}{BC}$ Substitute the known values.

45 [Challenging] • Target: Interior and exterior angles of polygons • Cross-Lesson: 8.01 Statistics & Probability

In the diagram, $ABCD$ is a trapezoid with $AB \parallel DC$. The diagonals AC and BD intersect at point E. Given that $AE = 3$ cm, $EC = 6$ cm, and $BE = 4$ cm, find the length of BD.

a. 10 cm

b. **12 cm** ✓

c. 16 cm

d. 18 cm

Since $AB \parallel DC$, we have that $\Delta ABE \sim \Delta CDE$ (by AAA similarity, where $\angle BAE = \angle DCE$ and $\angle ABE = \angle CDE$ are alternate interior angles, and $\angle AEB = \angle CED$ are vertically opposite angles). Therefore, the ratio of corresponding sides is equal: $\frac{AE}{CE} = \frac{BE}{DE} = \frac{AB}{CD}$. We are given $AE = 3$ cm, $EC = 6$ cm, and $BE = 4$ cm. Plugging these values into the proportion, we get: $\frac{3}{6} = \frac{4}{DE}$ This simplifies to.

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